Some of the first astronomers to see Venus through the telescope immediately recognized that they were examining a world enshrouded by clouds. The clouds, we now know, are droplets of concentrated sulfuric acid, stained yellow by a little elemental sulfur. They lie high above the ground. In ordinary visible light there's no hint of what this planet's surface, some 50 kilometers below the cloud tops, is like, and for centuries the best we had were wild guesses.

You might conjecture that if we could take a much finer look there might be breaks in the clouds, revealing day by day, in bits and pieces, the mysterious surface ordinarily hidden from our view. Then the time of guesses would be over. The Earth is on average half cloud-covered. In the early days of Venus exploration, we saw no reason that Venus should be 100 percent overcast. If instead it was only 90 percent, or even 99 percent, cloud-covered, the transient patches of clearing might tell us much.

In 1960 and 1961, *Mariner*s *1* and *2*, the first American spacecraft designed to visit Venus, were being prepared. There were those, like me, who thought the ships should carry video cameras so they could radio pictures back to Earth. The same technology would be used a few years later when *Rangers 7, 8*, and *9* would photograph the Moon on the way to their crash landings—the last making a bull's-eye in the crater Alphonsus. But time was short for the Venus mission, and cameras were heavy. There were those who maintained that cameras weren't really scientific instruments, but rather catch-as-catch-can, razzle-dazzle, pandering to the public, and unable to answer a single straightforward, well-posed scientific question. I thought myself that whether there are breaks in the clouds was one such question. I argued that cameras could also answer questions that we were too dumb even to pose. I argued that pictures were the only way to show the public—who were, after all, footing the bill—the excitement of robotic missions. At any rate, no camera was flown, and subsequent missions have, for this particular world, at least partly vindicated that judgment: Even at high resolution from close flybys, in visible light it turns out there are no